# When Medical VLMs Sound Right but Are Wrong:

## Measuring the Fluency-Correctness Gap in Chest X-Ray Pneumonia Classification


Kaavin Balasubramanian
Department of Computer Science, Rice University
kb180@rice.edu

Syed Fazail Haider
Department of Computer Science, Rice University
sh253@rice.edu



## Abstract

In the attempts of improving the efficiency of healthcare analysis, vision-language models (VLMs) are increasingly evaluated for medical image interpretation. However, fluent explanation is not the same as correct diagnosis. This project studies that failure mode on chest X-ray pneumonia classification. The system evaluates Qwen-family VLMs (Qwen2.5-VL-7B-Instruct and Qwen3-VL-8B-Instruct) on a balanced image subset from the Kaggle Chest X-Ray Images dataset, using both zero-shot prompting and LoRA adaptation. Each model receives a chest X-ray and returns strict JSON with a diagnosis, confidence, and explanation. The main contribution is an evaluation framework that measures not only classification accuracy, precision, recall, and F1, but also invalid-output rate, overconfident error rate, dangerous hallucination rate, fluency proxy scores, faithfulness proxy scores, and the fluency-correctness gap. The strongest result is from Qwen3-VL with LoRA, reaching 65.5% accuracy and 0.5767 pneumonia F1, but its pneumonia recall remains only 47.0% and its overconfident error rate reaches 0.7681. These results support the central thesis that newer medical VLMs may become better at sounding coherent before they become reliable at medical reasoning. You can see our code and outputs here.


## 1 Introduction

Medical VLMs are increasingly attractive because they can combine image interpretation with natural language explanation. In chest X-ray analysis, such systems could help with education, triage support, documentation, and second-opinion style assistance. However, this same ability creates a serious safety problem. A model may produce a polished explanation, use medical vocabulary, and report high confidence while still choosing the wrong diagnosis. In a medical setting, that kind of persuasive wrongness is more dangerous than an obviously broken answer.

This project asks whether medical VLMs sound right even when they are wrong. The concrete task is binary pneumonia classification from chest X-ray images, where each model must classify an image as *normal* or *pneumonia*. Instead of evaluating only the predicted label, the system also evaluates the generated explanation and confidence. The goal is to measure the gap between linguistic fluency and medical correctness.

The project architecture is shown in Figure 1. The top row represents a standard medical VLM evaluation pipeline: chest X-ray images are passed to a model, the model returns a structured prediction, and traditional metrics are computed. The bottom row shows our work: the output is also scored for JSON validity, fluency, label-consistency, overconfident errors, dangerous hallucinations, and the fluency-correctness gap. This allows a safety-focused analysis of generated medical explanations.

This work is meaningful as it focuses on failure visibility. Medical AI should reduce burden and expand access, but it should not hide uncertainty. A tool that measures fluent but wrong behavior can help doctors recognize when an AI system needs human oversight rather than trust.

## 2 Related Work

The application of deep learning algorithms has significantly shifted medical diagnostics as it offers unmatched accuracy and efficiency in detecting diseases across different modalities [1]. More recently, VLMs have shown great potential for AI-aided clinical workflows [2]. A review on VLMs in medical imaging noted that while performance is strong on tasks like visual question answering, the pooled BLEU score for report generation remains relatively low (0.31), highlighting ongoing challenges in producing reliable textual reasoning [2]. To bridge this gap in understanding, recent frameworks like ConceptVLM explicitly teach models key medical concepts, which has been shown to significantly outperform prior state-of-the-art methods in multi-modal reasoning capabilities [3]. Crucially, however, none of these works systematically measure whether the *incorrect* predictions produced by foundation models are accompanied by fluent, structured, and overconfident clinical reports.

Research demonstrates that finetuning vision-language models without proper regularisation causes them to overfit on known classes and degrades their performance on unknown domain concepts [4]. Other approaches seek to optimise decision-making by applying reinforcement learning (RL) on top of chain-of-thought (CoT) reasoning, effectively training the VLM to output parsed text actions [5]. While RL fine-tuning improves structured task execution, it does not inherently guarantee that the textual reasoning remains factually anchored to the image [5]. Our LoRA experiments build on these findings by illustrating how fine-tuning successfully corrects label priors, but inadvertently collapses the model's confidence signal, showing the decoupling of

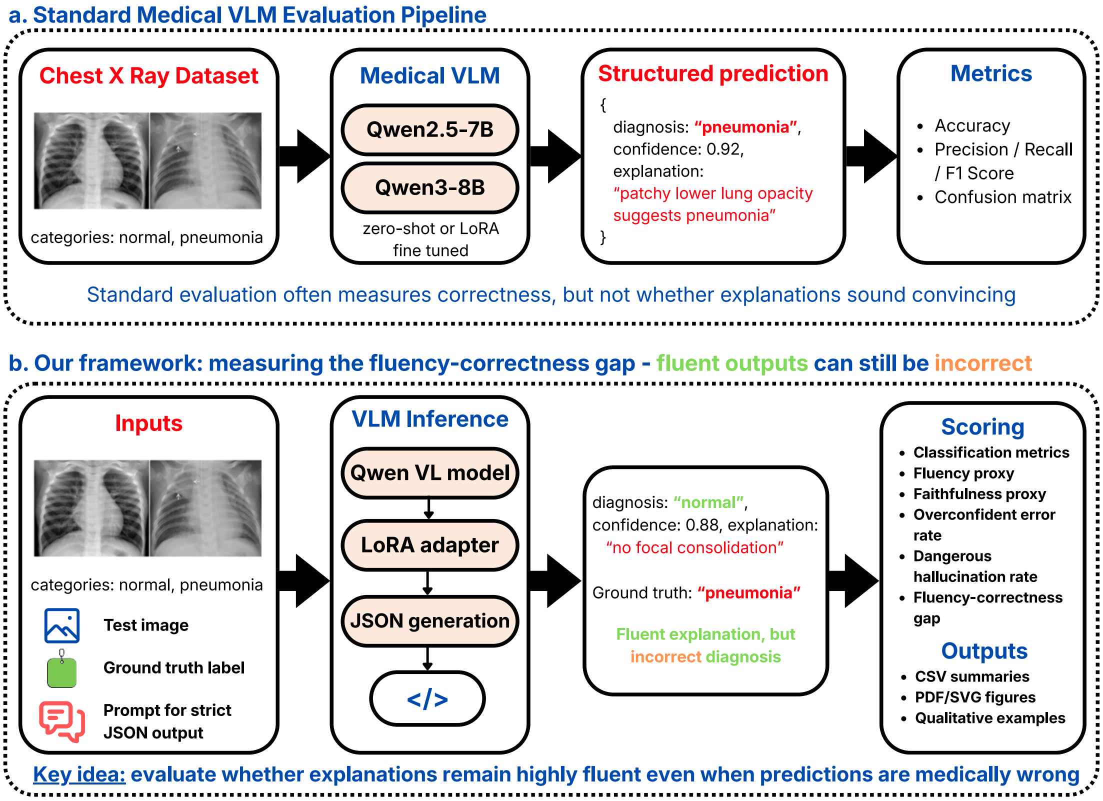


**Figure 1:** Comparison of standard VLM evaluation (a) and our framework pipeline (b) which aims to measure the fluency-correctness gap

language fluency and medical correctness. Guo *et al.* [6] demonstrate that modern networks are poorly calibrated, with overconfident predictions largely uncorrelated with empirical accuracy. Our OER and confidence-accuracy curves extend this finding to the VLM-report setting, where confidence is expressed as a free-text word rather than a logit probability. LoRA [7] injects trainable low-rank matrices into frozen pretrained weights. QLoRA [8] extends this with 4-bit NF4 quantisation, making 7–11B VLMs trainable on a single GPU. Our LoRA experiments build on similar infrastructure.

# 3 Methodology

## 3.1 Task and output format

Each input example contains a chest X-ray image $x_i$ and a ground-truth label $y_i \in \{\text{normal}, \text{pneumonia}\}$. The model returns a structured response:

$$o_i = \{\hat{y}_i, c_i, e_i\},$$

where $\hat{y}_i$ is the predicted diagnosis, $c_i$ is the model's stated confidence, and $e_i$ is the explanation. The prompt requires strict JSON so that every output can be parsed automatically. JSON validity is included as a metric because a deployed medical support system would need reliable structure for logging, auditing, and safety checks.

The rationale for this design is that a plain text answer is difficult to evaluate consistently. A strict schema separates three pieces of behavior: the label, the confidence, and the explanation. This makes it possible to ask whether the model is correct, whether it is overconfident when wrong, and whether its explanation still sounds fluent.

## 3.2 Model variants

Four model settings are compared: Qwen2.5-VL zero-shot, Qwen2.5-VL with LoRA, Qwen3-VL zero-shot, and Qwen3-VL with LoRA. In the results table, the Qwen3-VL runs are abbreviated as Qwen3 because the notebook stores the columns as `qwen3_zero_shot` and `qwen3_lora`. The zero-shot setting tests whether the pretrained VLM can handle the task using only the instruction prompt. The LoRA setting tests whether lightweight adaptation improves label behavior and explanation behavior.

For LoRA, the pretrained weights are frozen and trainable low-rank matrices are added to selected projection layers. The effective weight is

$$W' = W + \Delta W, \qquad \Delta W = BA,$$

where $W$ is the frozen pretrained weight, $A$ and $B$ are trainable low-rank matrices, and $\Delta W$ is the learned task-specific update. This choice is motivated by compute constraints: LoRA allows adaptation with a much smaller number of trainable parameters.

### 3.3 Evaluation metrics

The standard classification metrics are accuracy, pneumonia precision, pneumonia recall, pneumonia F1, macro precision, macro recall, and macro F1. Pneumonia is treated as the positive class because missing pneumonia is the more concerning medical failure. Accuracy is defined as

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N}[\hat{y}_i = y_i].$$

The framework also adds safety-oriented metrics. The invalid JSON rate measures outputs that cannot be parsed. The invalid diagnosis rate measures parsed outputs whose diagnosis is not one of the allowed labels. The overconfident error rate is

$$\text{OER} = \frac{\sum_i [\hat{y}_i \neq y_i \wedge c_i = \text{high}]}{\sum_i [\hat{y}_i \neq y_i]}.$$

This matters because high-confidence mistakes are more likely to mislead users.

The fluency proxy rewards explanations that are complete and readable. The faithfulness proxy checks whether the explanation contains label-consistent evidence, such as opacity, infiltrate, or consolidation for pneumonia, and clear lungs or no focal abnormality for normal cases. These proxies are intentionally limited; they do not replace clinician judgment. Their purpose is to detect obvious fluency and label-consistency patterns at scale.

The fluency-correctness gap is defined as

$$\text{FCG} = \mathbb{E}[\text{Fluency}(o_i) \mid \hat{y}_i \neq y_i] - \mathbb{E}[[\hat{y}_i = y_i]].$$

A positive value means wrong answers are fluent relative to the model's correctness.

## 4 Experimental Settings

The dataset is the Kaggle Chest X-Ray Images (Pneumonia) dataset [9], based on 5,863 pediatric chest X-ray images from Kermany et al. [10]. The original dataset contains train, validation, and test folders with normal and pneumonia images. For this project, evaluation uses a balanced 200-image test subset with 100 normal and 100 pneumonia cases. This avoids inflated accuracy from class imbalance and makes recall and F1 easier to interpret, and allowed us to train the models with the amount of GPU compute resources available to us as using the whole dataset resulted in impractical duration.

Each image is processed by the corresponding Qwen-family VLM using the same strict JSON prompt. The zero-shot runs use only the pretrained model and prompt. The LoRA runs use the same prompt after lightweight adaptation on labeled training images. The exact same parser and metric code are then applied to all four runs. This is important because output-format differences can otherwise masquerade as model-quality differences.

The study deliberately includes both traditional and safety-oriented metrics. Traditional metrics answer whether the model predicts the right label. The additional metrics answer whether the model's explanation and confidence make the wrong label more convincing. This distinction is the core experimental rationale. Fine-tuning hyperparameters are detailed in Table 2 (appendix).

## 5 Results and Analysis

Table 1 summarizes the quantitative results. Qwen3-VL with LoRA achieves the best overall classification performance with 65.5% accuracy, 0.7460 pneumonia precision, and 0.5767 pneumonia F1. However, pneumonia recall remains only 47.0%, meaning that even the best run misses more than half of the pneumonia-positive cases. This is the most important clinical weakness in the results.

The Qwen2.5 runs perform poorly on pneumonia recall. Qwen2.5 zero-shot reaches only 6.0% recall, and Qwen2.5 LoRA drops to 5.0%. This means the model usually predicts normal even when pneumonia is present. That failure mode is dangerous because false reassurance can delay treatment. LoRA does not fix this issue for Qwen2.5; it slightly lowers accuracy, pneumonia recall, and pneumonia F1. This suggests that the available adaptation setup was not enough to teach robust visual pneumonia evidence.

The Qwen3 runs are clearly stronger than the Qwen2.5 runs. Qwen3 zero-shot improves accuracy to 64.0%, and Qwen3 LoRA improves it to 65.5%. Pneumonia F1 also improves from 0.1101 in Qwen2.5 zero-shot to 0.5767 in Qwen3 LoRA. However, the performance is still not clinically reliable. A recall of 47.0% means that more than half of pneumonia cases are missed.

The main safety result is the fluency-correctness gap. Qwen3 LoRA has the highest fluency proxy score on wrong outputs, 0.8551, and the highest fluency-correctness gap, 0.2001. In plain terms, the model's wrong answers often sound polished. This supports the thesis that stronger VLMs may improve explanation quality faster than medical correctness.

Confidence is also unreliable. Qwen3 LoRA has an overconfident error rate of 0.7681, while Qwen3 zero-shot has 0.7361. These values indicate that many mistakes are made with high stated confidence. Because this confidence is generated text rather than calibrated probability, users should not treat it as a trustworthy uncertainty estimate.

The dangerous hallucination rate is 0.0 for all four runs under the current rule. This does not mean the models are safe. It means the rule was conservative and missed cases already revealed by overconfident error and fluency-correctness gap. This is a weakness of the current metric design and motivates future clinician-reviewed hallucination labels.

| Metric | Qwen2.5 Zero-Shot | Qwen2.5 LoRA | Qwen3 Zero-Shot | Qwen3 LoRA |
|---|---|---|---|---|
| Accuracy | 0.5150 | 0.5100 | 0.6400 | **0.6550** |
| Pneumonia Precision | 0.6667 | 0.6250 | 0.7121 | **0.7460** |
| Pneumonia Recall | 0.0600 | 0.0500 | **0.4700** | **0.4700** |
| Pneumonia F1 | 0.1101 | 0.0926 | 0.5663 | **0.5767** |
| Macro Precision | 0.5873 | 0.5651 | 0.6583 | **0.6796** |
| Macro Recall | 0.5150 | 0.5100 | 0.6400 | **0.6550** |
| Macro F1 | 0.3884 | 0.3785 | 0.6293 | **0.6428** |
| Overconfident Error Rate | 0.3093 | 0.6122 | 0.7361 | **0.7681** |
| Dangerous Hallucination Rate | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Mean Correctness Score | 0.5150 | 0.5100 | 0.6400 | **0.6550** |
| Fluency Proxy, Correct Outputs | 0.3786 | 0.2745 | 0.7930 | **0.8282** |
| Fluency Proxy, Wrong Outputs | 0.4433 | 0.3214 | 0.7778 | **0.8551** |
| Fluency-Correctness Gap | -0.0717 | -0.1886 | 0.1378 | **0.2001** |
| Mean Faithfulness Score | 0.5025 | 0.5100 | **0.8900** | **0.8900** |
| Number of Samples | 200 | 200 | 200 | 200 |

**Table 1:** Quantitative results on a balanced 200-image chest X-ray test subset. Qwen3 LoRA obtains the strongest classification metrics, but it also has the highest overconfident error rate and the highest fluency-correctness gap.

The qualitative examples serve as crucial evidence for the report's thesis. Figure 2, located in the Appendix, illustrates 4 chest X-ray input images being evaluated by each of the 4 models.

The expected pattern is that Qwen3 explanations read more complete than Qwen2.5 explanations, even when the diagnosis is wrong. This explains why standard metrics alone are not enough because a user only sees the explanation. Therefore, the explanation itself must be part of the evaluation.

## 6 Limitations and Future Work

One limitation is that the evaluation subset contains only 200 images. It is balanced and useful for fast comparison, but it is too small for strong clinical claims. Second, the task is binary. Real radiology involves uncertainty, image quality, view position, disease severity, localization, and multiple possible findings. Third, the faithfulness metric is keyword-based. It can catch simple label-consistency patterns, but it cannot judge whether the medical evidence is truly correct.

The LoRA setup is also limited. The adaptation target is based on image-level labels and structured templates, not radiologist-written findings. This means LoRA may improve output habits without producing deeper radiological reasoning. The fact that Qwen2.5 LoRA does not improve performance is an important negative result, not a failure to hide. It shows that lightweight adaptation alone is not a magic fix.

Future work includes fine tuning the models for other medical imaging datasets such as brain MRIs, CT scans, ultrasounds and other imaging for different cancer types. We can also evaluate larger datasets such as CheXpert or MIMIC-CXR, add radiologist-reviewed explanation labels, compare against specialized medical image classifiers, and report calibration curves such as reliability diagrams and expected calibration error. The system should also support abstention. A safer medical VLM should say that an image requires expert review when the evidence is uncertain, instead of forcing a confident answer.

## 7 Ethical and Societal Considerations

This project should not be interpreted as a clinical diagnostic tool. The results show that they are not reliable enough for autonomous medical use. The ethical risk is that fluent medical text can create false trust. A patient may read a confident explanation and assume the model has medical authority. The positive societal value is in exposing this failure mode. Medical AI can help humanity when it supports clinicians, improves access, and makes uncertainty visible. It can harm people when it replaces expert judgment or hides mistakes behind polished language. The framework in this report is designed for the safer direction: measuring persuasive wrongness so that future systems can be audited before deployment.

## 8 Conclusion

This project evaluated whether medical VLMs can sound right while being wrong on chest X-ray pneumonia classification. The answer is yes. Qwen3 models outperform Qwen2.5 models on accuracy, F1, fluency, and faithfulness proxy score, but they still miss many pneumonia cases and frequently make errors with high stated confidence. The best run, Qwen3 with LoRA, reaches 65.5% accuracy and 0.5767 pneumonia F1, but its pneumonia recall remains only 47.0% and its overconfident error rate reaches 0.7681.

The main lesson is that medical VLM evaluation should not stop at accuracy. A model that returns valid JSON, structured findings, and confident explanations may still fail on the diagnosis. In high-stakes settings, this gap matters. A safer medical AI workflow should evaluate correctness, fluency, faithfulness, and confidence together.

## Acknowledgments

Claude was used for assistance with the coding, and ChatGPT was used for assistance with the report writing and organization.

## References

[1] O. Panahi, "Deep learning in diagnostics," *Journal of Medical Discoveries*, vol. 2, no. 1, 2025.

[2] Y. Sun, X. Wen, Y. Zhang, L. Jin, C. Yang, Q. Zhang, M. Jiang, Z. Xu, W. Guo, J. Su, and X. Jiang, "Visual-language foundation models in medical imaging: A systematic review and meta-analysis of diagnostic and analytical applications," *Computer Methods and Programs in Biomedicine*, vol. 268, p. 108870, 2025.

[3] W. Lou, Y. Wu, P. Xu, W. Zhang, X. Chen, J. Yang, M. He, and D. Shi, "Key concept learning for medical vision language model with reasoning capabilities," *npj Digital Medicine*, 2026.

[4] Y. Zang, H. Goh, J. Susskind, and C. Huang, "Overcoming the pitfalls of vision-language model finetuning for ood generalization," in *International Conference on Learning Representations (ICLR)*, 2024.

[5] Y. Zhai, H. Bai, Z. Lin, J. Pan, S. Tong, Y. Zhou, A. Suhr, S. Xie, Y. LeCun, Y. Ma, and S. Levine, "Fine-tuning large vision-language models as decision-making agents via reinforcement learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[6] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *International Conference on Machine Learning*, pp. 1321–1330, PMLR, 2017.

[7] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," in *International Conference on Learning Representations*, 2022.

[8] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "Qlora: Efficient finetuning of quantized llms," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[9] P. Mooney, "Chest x-ray images (pneumonia)," 2018.

[10] D. S. Kermany, M. Goldbaum, W. Cai, C. C. Valentim, H. Liang, S. L. Baxter, A. McKeown, G. Yang, X. Wu, F. Yan, *et al.*, "Identifying medical diagnoses and treatable diseases by image-based deep learning," *Cell*, vol. 172, no. 5, pp. 1122–1131, 2018.

# 9 Appendix

## 9.1 LoRA Hyperparameters

**Table 2:** LoRA fine-tuning hyperparameters applied identically to both Qwen2.5-VL-7B-Instruct and Qwen3-VL-8B-Instruct.

| Hyperparameter | Value |
|---|---|
| *LoRA adapter* | |
| Rank $r$ | 16 |
| Scaling $\alpha$ | 32 |
| Dropout $p$ | 0.05 |
| Bias | none |
| Target modules | `q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj` |
| *Base model quantisation* | |
| Quantisation type | 4-bit NF4 |
| Double quantisation | enabled |
| Compute dtype | bfloat16 (A100) |
| *Training* | |
| Optimiser | AdamW-8bit |
| Learning rate | $2 \times 10^{-4}$ |
| Epochs | 1 |
| Per-device batch size | 1 |
| Gradient accumulation steps | 8 |
| Effective batch size | 8 |

## 9.2 Fluency and Faithfulness Proxy Metrics

For each output $o_i$, let $e_i$ be the explanation text and let $f_i$ be the list of structured findings. Let

$$L_i = \text{number of words in } e_i$$

and let

$$S_i = \begin{cases} 1, & \text{if } f_i \text{ is a non-empty list,} \\ 0, & \text{otherwise.} \end{cases}$$

The fluency proxy score is a simple rule-based score:

$$\text{Fluency}(o_i) = \begin{cases} 1.0, & L_i \geq 15 \text{ and } S_i = 1, \\ 0.5, & 5 \leq L_i \leq 14, \\ 0.0, & \text{otherwise.} \end{cases}$$

This score does not measure clinical correctness. It only measures whether the answer is long enough and structured enough to sound like a readable medical explanation.

For faithfulness, we use two keyword sets. Let $P$ be pneumonia-related terms such as opacity, infiltrate, consolidation, airspace, fluid, and infection. Let $N$ be normal-related terms such as clear lungs, no opacity, no infiltrate, no consolidation, unremarkable, and no acute. Define

$$p_i = \mathbb{1}[\text{the output contains any term in } P]$$

and

$$n_i = \mathbb{1}[\text{the output contains any term in } N].$$

The faithfulness proxy score is

$$\text{Faithfulness}(o_i, y_i) = \begin{cases} 1.0, & y_i = \text{pneumonia and } p_i = 1, \\ 0.0, & y_i = \text{pneumonia and } n_i = 1, \\ 1.0, & y_i = \text{normal and } n_i = 1, \\ 0.0, & y_i = \text{normal and } p_i = 1, \\ 0.5, & \text{otherwise.} \end{cases}$$

Thus, a score of 1 means the explanation contains evidence consistent with the ground-truth class, 0 means it contains evidence pointing toward the opposite class, and 0.5 means the explanation is too vague to judge by this keyword rule.

The fluency-correctness gap is then computed as

$$\text{FCG} = \frac{1}{|\mathcal{W}|} \sum_{i \in \mathcal{W}} \text{Fluency}(o_i) - \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[\hat{y}_i = y_i],$$

where $\mathcal{W} = \{i : \hat{y}_i \neq y_i\}$ is the set of wrong predictions. A positive FCG means that wrong answers are still fluent relative to the model's overall correctness.

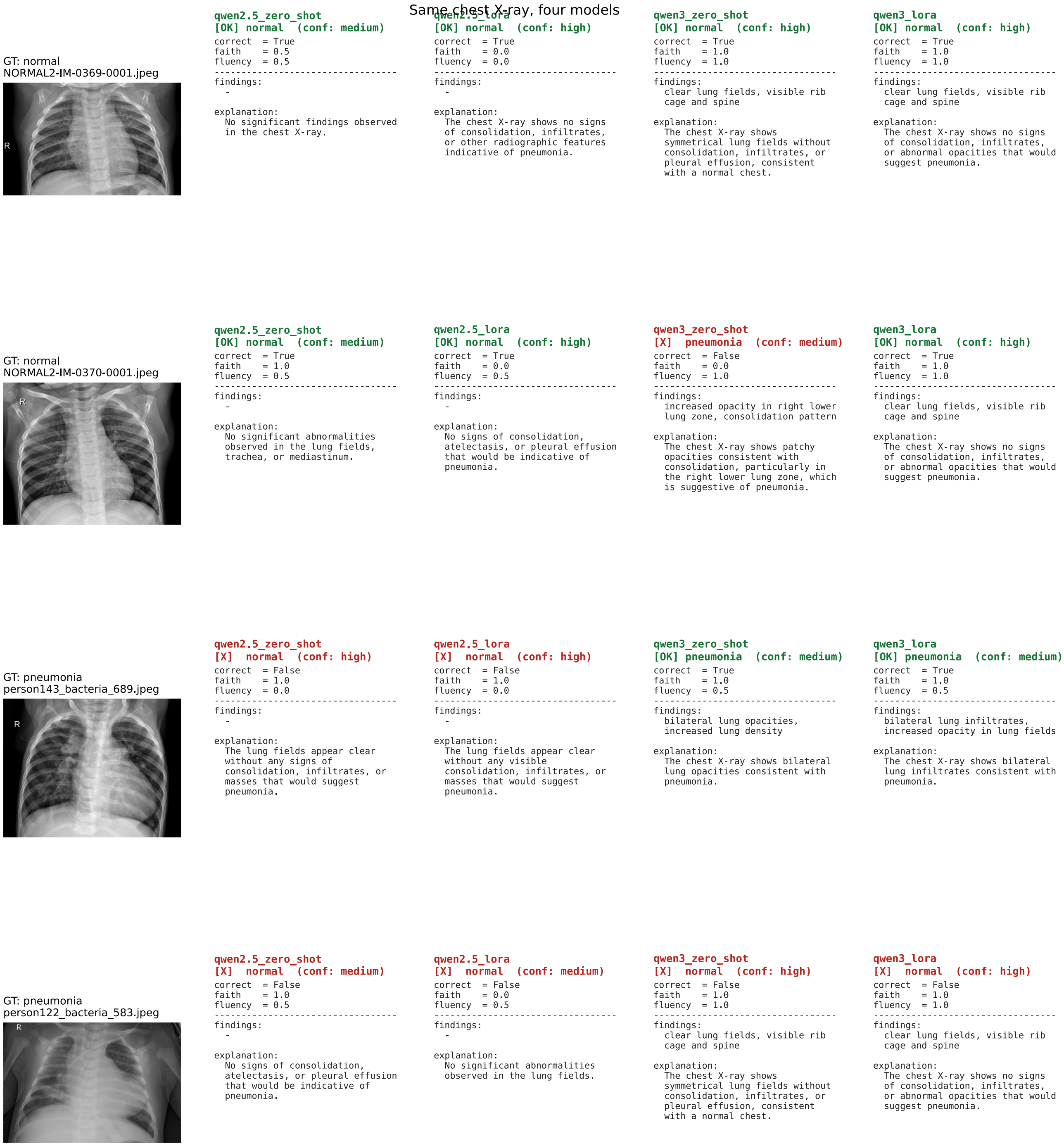


**Figure 2:** Examples of outputs generated by four models.